

Rajashekhar S
Sr. Quality Assurance Engineer
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PROFESSIONAL SUMMARY:

- Over 9 years of expertise in **software testing**, including **manual and automated testing** of complex applications across various domains. Skilled in **designing, implementing, and executing test strategies** to ensure software quality and reliability.
- Strong knowledge of **software testing methodologies and best practices**, including **Software Development Life Cycle (SDLC)**, **Software Testing Life Cycle (STLC)**, and various testing approaches like **Functional, Regression, Integration, System, End-to-End, Performance, Security, and Exploratory Testing**.
- Proficient in **automation testing frameworks and tools** including **Selenium WebDriver, Protractor, Cucumber, Robot Framework, JMeter, Gatling, and SOAP UI** and Experienced in developing reusable automation scripts and frameworks for **web, API, ETL and database testing**.
- Expert in **API and Web Services Testing** using **SOAP UI, Postman, REST Assured, and Python requests library**. Skilled in testing **REST and SOAP-based web services**, handling **XML/WSDL**, and automating API validation processes.
- Strong **programming and scripting** experience with **Python, Java, and JavaScript**, used for developing automation test scripts, integrating with **CI/CD pipelines**, and enhancing test coverage. Proficient in **Python PyTest, requests, Selenium WebDriver, and boto3 libraries**.
- Hands-on experience with **ETL testing**, including working with **SSIS pipelines**, processing diverse file formats, and performing complex **SQL-based data validations** in **MS SQL Server**.
- Skilled in **AWS and Azure cloud testing**, including validating services such as **AWS Lambda, API Gateway, S3, IAM, CloudWatch, SNS, SQS, EC2, RDS**, and **Azure DevOps tools like Azure Boards, Pipelines, Repos, Test Plans, and Artifacts**.
- Experience in **continuous integration and deployment (CI/CD) workflows**, working with tools like **Jenkins, Maven, SonarQube, Git, and Azure DevOps** to automate test execution and improve software delivery efficiency.
- Expert in **Selenium automation** with deep knowledge of **XPath, CSS Selectors**, and object identification techniques to handle **dynamic UI elements, iframes, pop-ups, dropdowns, radio buttons**, and other complex web components.
- Strong knowledge of **SQL and database testing**, including writing complex queries using **joins, aggregations, window functions**, and performing data validation across multiple database systems.
- Experience in **load and performance testing** using **Apache JMeter and Gatling**, ensuring the stability and scalability of web applications and APIs.
- Experienced in **Agile and DevOps methodologies**, including **Scrum, Kanban, Test-Driven Development (TDD), and Model-Driven Development (MDD)**. Active collaboration with development and product teams to ensure seamless testing and deployment.
- Proficient in **test management and documentation**, including creating **test plans, test cases, Requirements Traceability Matrices (RTM), defect reports, and post-deployment documentation**.
- Worked with **Tableau** for **data visualization**, connecting to various data sources and creating insightful dashboards and reports.
- Strong **analytical, problem-solving, and troubleshooting skills**, with a keen eye for identifying and resolving software defects efficiently.
- Excellent **communication and collaboration skills**, ensuring effective coordination between cross-functional teams to meet project deadlines and quality standards.

TECHNICAL SKILLS:

Programming Languages	Python, SQL, Java, JS (Java script)
Test Tools	Postman, Soap UI, Apache JMeter, Appium Server & Inspector

Test Build & Integration Tools	Maven, Jenkins, SonarQube, Team city
Test Frameworks	PyTest, PyMock, Junit, TestNg
Databases	Oracle, Microsoft SQL Server, Postgres, MYSQL, MongoDB
Cloud Services	AWS Lambda, API Gateway, SNS, CloudWatch, IAM, S3, EC2, RDS
Markup Languages	HTML, XML, XPath
Operating Systems	Windows, Mac, Linux
Defect Tracking tools	JIRA, HPQC/ ALM, MTM, TFS
MS Office Tools	Outlook, Word, Excel, PowerPoint.
Project Methodologies	Agile-Scrum, Waterfall model

PROFESSIONAL EXPERIENCE:

Meta/Facebook, AZ | Remote (Contract)
Role: Sr. QA Engineer

Feb 2022 – Present

Key Responsibilities:

- **Designed and implemented manual and automated test suites** to validate software functionality, performance, and reliability across various systems and platforms. Collaborated with business users and SMEs to define test plans, scenarios, and acceptance criteria.
- Performed **functional, integration, regression, user acceptance (UAT), system, load, and black-box testing**, ensuring software met business requirements and performance expectations.
- Validated multiple **AWS services including Lambda, API Gateway, SNS, SQS, CloudWatch, and RDS** using **Postman** for manual API testing and **Python requests library** for automation. Analyzed **CloudWatch logs** and monitored system alerts via **AWS SNS**.
- Conducted **ETL pipeline testing** by processing **CSV, Avro, and Parquet files** through **AWS Glue jobs**, ensuring accurate transformation and loading into **AWS Redshift databases**. Applied **PySpark** and **SQL queries** to verify data integrity.
- Developed **SQL queries** leveraging **joins, aggregations, and window functions** to compare source files with destination databases and validate schema changes across on-premise and cloud environments using **Python Pandas**.
- **Created and maintained Python-based automation frameworks** using **PyTest**, incorporating parameterization, grouping, and priority decorators for structured test execution.
- **Built front-end automation test scripts** using **Cypress JS**, performing UI validations, taking screenshots, and working with hooks, fixtures, and selector playground to identify elements efficiently.
- **Analyzed test failures on AWS CloudWatch**, debugging logs, and ensuring **SNS-based alert notifications** were triggered for critical failures.
- **Conducted performance testing of REST APIs** using **Apache JMeter**, generating detailed performance reports and performing distributed load testing using a **JMeter master-slave architecture** across multiple machines.
- **Scheduled automated JMeter test executions in Jenkins**, comparing test results against baseline performance metrics to track system behavior over time.
- **Validated hardware device communication protocols, including I2C, SPI, and CAN**, by analyzing hexdump data packets and debugging communication failures on embedded systems.
- **Managed Jenkins-based regression test suites**, scheduling periodic test runs and monitoring failures to ensure pre-release quality assurance.
- **Conducted requirement analysis and user story validation in Agile Scrum environments**, collaborating with product owners and business analysts to ensure full test coverage.
- **Provided sprint story estimations**, actively participated in sprint planning meetings, and contributed to defining test scope and priorities.
- **Designed and implemented automated test suites** using **Python, PyTest, and Page Object Model (POM)** for scalable and maintainable test coverage.

- **Performed gRPC microservices testing for GoLang-based services**, utilizing **BloomRPC** to validate protobuf messages exchanged between systems.
- **Maintained defect tracking and resolution workflows in JIRA**, documenting issues and coordinating with developers to ensure timely fixes and retesting.
- **Executed SQL-based validations on Oracle databases**, manually querying datasets to validate business logic and supporting data-driven testing methodologies.
- **Verified AWS Glue job executions**, ensuring seamless data ingestion into **AWS DynamoDB** by reading from **S3 buckets** and inserting into **AWS Redshift**.
- **Used Postman for API testing**, sending RESTful requests, validating XML and JSON responses, and automating end-to-end API workflows.
- **Integrated test suites with Jenkins CI/CD**, enabling automated test executions on code commits and maintaining software quality across deployment cycles.
- **Conducted high-performance backend service testing for financial trading applications** using **JMeter**, ensuring low-latency execution of trading transactions.
- **Tested AWS API endpoints processed through Lambda functions**, ensuring accurate responses and seamless integration between cloud-based services.

Environment: AWS (Lambda, API Gateway, SQS, SNS, CloudWatch, Glue, Redshift), Python (PyTest, Pandas, PySpark), Cypress JS, JMeter, gRPC (BloomRPC), Jenkins, Oracle, I2C/SPI/CAN protocols

Morgan Stanley, AZ |

Sep 2018 – Jan 2022

Role: Sr. QA Automation Engineer

Responsibilities:

- **Engaged in analyzing business requirements and technical specifications** to develop comprehensive test plans and detailed test cases, ensuring complete test coverage of the application.
- **Followed a Test-Driven Development (TDD) approach**, implementing test cases using **PyTest**, **Selenium**, and **PyMock** for robust and efficient automation testing.
- **Collaborated with cross-functional teams** during requirement reviews to assess modification and enhancement requests, contributing to seamless application improvements.
- **Designed and executed automation frameworks for web, mobile, and API testing**, utilizing **Selenium WebDriver**, **Appium**, and **REST API automation with Postman and custom Python scripts**.
- **Developed and implemented automation test scripts in Java** for black-box and white-box testing, covering functional, non-functional, and performance aspects of the application.
- **Validated RESTful APIs** by leveraging **Postman and custom Python scripts**, ensuring seamless integration with backend services. Enhanced API automation using **PyTest and Jenkins pipelines** for continuous execution.
- **Built automation frameworks using Cucumber and Selenium WebDriver**, incorporating data-driven and behavior-driven approaches (**BDD**) to enhance test scalability and efficiency.
- **Worked extensively on ETL pipeline testing**, processing various file formats (**CSV, Avro, Parquet**) and loading them into **MS SQL Server** using **Microsoft SSIS Packages**, ensuring accurate data transformation.
- **Developed and optimized SQL queries** with complex joins, aggregations, and window functions to verify data integrity across ETL pipelines and database systems.
- **Implemented performance testing strategies using JMeter**, measuring API response times, generating performance reports, and documenting results on **Confluence** pages for team-wide visibility.
- **Integrated automation scripts with Jenkins**, configuring **CI/CD pipelines** to schedule test execution and ensure early defect detection in nightly batch runs. Developed **Groovy scripts** to enhance Jenkins pipeline functionality.

- **Utilized Appium for mobile testing**, automating test cases for **iOS and Android applications** by configuring **Appium Server and Inspector** to connect with device simulators.
- **Developed comprehensive dashboards and reports in Tableau**, visualizing key business metrics using bar graphs, pie charts, and interactive visualizations to support data-driven decision-making.
- **Executed UI and cross-browser testing** using **Selenium WebDriver** to validate application compatibility across different browsers like **Chrome, Firefox, and Internet Explorer**.
- **Designed and implemented a test framework** using **Page Object Model (POM), TestNG, and Cucumber**, enhancing test organization and maintainability. Parameterized test cases using **Data Provider** annotation in TestNG.
- **Tested SOAP and RESTful web services** using **SOAP UI and Postman**, ensuring API schema compliance, validating responses, and automating workflows through assertion methods.
- **Conducted backend database validations**, checking data consistency by extracting and verifying datasets through **SQL queries in Oracle and MS SQL Server**.
- **Documented best practices for test execution, maintenance, and enhancements**, ensuring future test scalability and knowledge transfer for QA teams.

Environment: Selenium WebDriver (Java/Python), Appium, Cucumber, TestNG, Postman, SOAP UI, MS SQL Server, SSIS, Oracle, Jenkins, Tableau

Almac Health, PA |

Feb 2017 – Aug 2018

Role: QA Automation Engineer

Responsibilities:

- **Took ownership of preparing test cycle reports**, analyzing defect metrics, and implementing improvement actions for subsequent iterations to enhance test efficiency.
- **Conducted extensive ETL testing**, ensuring accurate extraction, transformation, and loading of dimensions and facts for user and tax data validation.
- **Developed automation scripts in Python** using **PyCharm IDE**, integrating libraries such as **requests, BeautifulSoup, and Kafka events** for backend testing.
- **Leveraged Python Pandas** for data comparison, loading structured files into data frames, and verifying accuracy between source and destination datasets.
- **Collaborated with cloud teams**, utilizing **AWS Lambda, S3, and Event Manager** to trigger and monitor ETL pipelines handling **CSV, Avro, and Parquet** data formats.
- **Built and executed UI automation tests** using **Cypress JS**, integrating them into **Jenkins pipelines** for scheduled regression runs, ensuring front-end consistency.
- **Enhanced test case robustness** by debugging **Cypress failures**, refining locators, and implementing retry mechanisms for improved reliability.
- **Crafted complex SQL queries** utilizing **inner/outer joins, aggregations, and window functions** to validate cross-system data integrity and performance.
- **Gained hands-on experience in testing embedded systems**, working with **ECU (Electronic Control Units), ADCs (Analog to Digital Converters), pressure sensors, and temperature sensors**.
- **Validated peripheral device communication over I2C and SPI protocols**, ensuring seamless data transfer on embedded system buses.
- **Utilized analytical tools** such as **MATLAB Simulink and bus analyzers** to examine hexadecimal messages and troubleshoot communication discrepancies.
- **Designed and executed automated validation for nearly 200 customer reports**, integrating test frameworks into **Jenkins pipelines** for scheduled runs.
- **Contributed to the development of a hybrid automation framework** tailored for **AWS cloud services**, optimizing test execution across distributed environments.
- **Implemented test cases within the Robot Framework**, structuring corresponding config and robot files to ensure comprehensive test coverage.
- **Worked within agile methodologies**, applying **Acceptance Test-Driven Development (TDD)** to accelerate release cycles while maintaining quality.
- **Played a key role in testing batch file processing**, leveraging **Python Spark applications and Apache Airflow** for workflow orchestration.

- **Ensured seamless data transformation**, executing **SQL queries** to verify proper ETL processing, and automating validation within **Jenkins pipelines**.
- **Performed A/B testing for SQL stored procedure changes**, analyzing performance impacts and ensuring optimization before deployment.
- **Collaborated closely with test coordinators, development teams, and business stakeholders** to refine test cases and achieve maximum coverage.
- **Created comprehensive test cases** covering end-to-end workflows, functional scenarios, and **UAT processes** aligned with business requirements.
- **Conducted post-deployment smoke testing**, validating system stability and performing sanity checks across production environments.
- **Developed JavaScript scripts in Postman** to automate **REST API validations**, improving efficiency in web services testing.
- **Maintained and optimized regression automation scripts**, ensuring end-to-end test execution aligned with evolving application changes.
- **Designed XML-based test suites** to streamline web application testing and ensure data-driven validation of backend functionalities.
- **Utilized JIRA for bug tracking**, collaborated with development teams for resolution, and kept stakeholders informed about defect status and progress.
- **Engaged proactively with cross-functional teams**, adapting to changing requirements, coordinating builds, and supporting releases for smooth deployments.

Environment: AWS (Lambda, S3), Python (Pandas, Robot Framework), Cypress JS, MATLAB Simulink, Apache Airflow, PySpark, I2C/SPI protocols

Divi's Laboratories, India |

May 2015 – Dec 2016

Role: QA Analyst

Responsibilities:

- **Actively participated in Agile testing methodologies**, attending daily **Scrum meetings**, and ensuring thorough testing of each sprint deliverable.
- **Crafted, reviewed, and executed test cases and scripts for Functional, Integration, System, and GUI testing**, aligning them with business and requirement specifications.
- **Developed and implemented automated test scripts** using **Selenium WebDriver**, significantly enhancing efficiency in functional testing.
- **Engineered test code in Java** using **Eclipse, WebDriver, and TestNG**, contributing to a robust and scalable automation framework.
- **Designed and built an advanced Test Automation Framework** using **Selenium WebDriver, TestNG, and Maven**, leveraging industry-leading design patterns.
- **Conducted Web Services testing with SOAP UI**, validating **WSDLs**, analyzing request-response cycles, and ensuring seamless API communication.
- **Defined comprehensive test strategies**, including test plans, high-level scenarios, and resource estimates, while maintaining weekly status reports.
- **Led test case walkthroughs**, ensuring consensus on correctness, completeness, and extensive coverage of critical business scenarios.
- **Created and maintained Hybrid and Keyword-driven Automation Frameworks**, improving test execution efficiency and reusability.
- **Executed diverse testing methodologies**, including **Functional, Regression, System, Integration, Security, GUI, Configuration, and UAT**.
- **Developed automation scripts** using **Selenium, Java, TestNG, and Cucumber** as part of an **Acceptance Test-Driven Development (ATDD)** approach.
- **Utilized Git and GitHub for version control**, maintaining seamless collaboration and code repository management.
- **Implemented a robust Cucumber-based Test Automation Framework** integrated with **Selenium WebDriver, TestNG, and Maven** for effective testing.

- Worked **extensively with APIs**, leveraging **Postman** to validate **CRUD operations** and ensure smooth backend interactions.
- Performed **cross-browser testing**, validating application compatibility for dynamic and static content using **HTML IDs and XPath in Selenium**.
- Integrated **automated tests into Continuous Integration pipelines**, executing nightly and need-based test runs using **Maven and Jenkins**.
- **Reported, tracked, and collaborated on software defects** using **JIRA**, engaging with developers to resolve technical issues efficiently.

Environment: Selenium WebDriver (Java), TestNG, Cucumber, SOAP UI, Postman, Jenkins, Git, Oracle, MS SQL, JIRA, Agile/Scrum, Functional and Regression testing, cross browser